TERMS OF REFERENCE

TRAINING AND DEVELOPMENT IN DATA-DRIVEN MATHEMATICAL MODELING FOR EARTHQUAKE EPICENTER ESTIMATION IN PRE-INSTRUMENTAL AND CONTEMPORARY SEISMIC EVENTS

I. INTRODUCTION

The Terms of Reference (ToR) define the implementation framework, operational structure, guiding principles, responsibilities, expected outputs, and monitoring mechanisms of the Visiting Expert Program (VEP) under the Human Resource Development Program (HRDP) of the Department of Science and Technology – Philippine Council for Industry, Energy, and Emerging Technology Research and Development (DOST–PCIEERD).

The ToR serves as the primary institutional and operational reference for the implementation of the project entitled:

"TRAINING AND DEVELOPMENT IN DATA-DRIVEN MATHEMATICAL MODELING FOR EARTHQUAKE EPICENTER ESTIMATION IN PRE-INSTRUMENTAL AND CONTEMPORARY SEISMIC EVENTS

Implemented through the Ateneo de Davao University Mathematics Department in coordination with the Visiting Expert, Dr. Jeffrey S. Perez of the Philippine Institute of Volcanology and Seismology (DOST–PHIVOLCS), the project is structured as an interdisciplinary capacity-building and knowledge-transfer initiative focused on earthquake interpretation, historical seismic analysis, uncertainty-aware reasoning, and exploratory data-driven scientific workflows relevant to intensity-based earthquake epicenter estimation.

The project recognizes that historical and contemporary earthquake interpretation involves complex observational, geological, mathematical, and computational considerations. In many cases, earthquake-related information is incomplete, spatially variable, uncertain, and dependent on both scientific judgment and contextual interpretation. In response, the program emphasizes the development of conceptual understanding, methodological awareness, interdisciplinary integration, and responsible scientific reasoning through expert-guided learning activities, discussions, mentoring sessions, and structured analytical engagement.

This ToR establishes the scope, implementation boundaries, and governance structure of the program to ensure that all project activities remain aligned with the objectives and policies of the Visiting Expert Program. The document further guides participant engagement, documentation procedures, reporting requirements, implementation workflows, and the intended educational and institutional outcomes of the project.

The program is designed primarily as a capacity-building and knowledge-absorption initiative rather than as a formal research and development implementation project under the VEP mechanism. While discussions may include conceptual, exploratory, and interdisciplinary perspectives involving computational systems, data-driven analysis, geospatial interpretation, uncertainty-aware workflows, and emerging analytical approaches in seismology, such activities are intended solely for educational, methodological, and research-readiness purposes within the scope of the program. Accordingly, the project does not aim to develop operational earthquake prediction systems, deployable hazard platforms, or validated forecasting technologies under the Visiting Expert Program. Any conceptual, analytical, or prototype-based outputs that may emerge during the training activities shall be understood as exploratory and educational outputs intended to support interdisciplinary learning, scientific discussion, and future research planning beyond the scope of the current VEP implementation.

The ToR also promotes institutional coherence and continuity by consolidating the project's implementation framework, session structure, expected outputs, and administrative processes into a unified operational document. All project participants, faculty members, trainees, and collaborating personnel are expected to observe the provisions, limitations, responsibilities, and implementation guidelines described herein throughout the duration of the program.

Through this framework, the project seeks to strengthen institutional competencies in seismology-informed earthquake interpretation, foster interdisciplinary scientific engagement, and support the long-term development of responsible and methodologically grounded academic initiatives related to earthquake analysis, historical seismicity, and disaster risk understanding.

II. PROJECT INFORMATION

A. Project Title

TRAINING AND DEVELOPMENT IN DATA-DRIVEN MATHEMATICAL MODELING FOR EARTHQUAKE EPICENTER ESTIMATION IN PRE-INSTRUMENTAL AND CONTEMPORARY SEISMIC EVENTS

This title shall serve as the official and authoritative project title for all implementation, reporting, monitoring, documentation, and administrative purposes related to the Visiting Expert Program under DOST–PCIEERD.

B. Implementing Institution

Ateneo de Davao University
Ateneo de Davao University Mathematics Department
Roxas Avenue, Poblacion District, Davao City, Philippines

The Ateneo de Davao University Mathematics Department shall serve as the primary implementing academic unit responsible for coordinating the overall implementation, participant engagement, institutional documentation, and administrative execution of the program in accordance with DOST–PCIEERD policies and institutional requirements.

C. Funding Agency

Department of Science and Technology – Philippine Council for Industry, Energy, and Emerging Technology Research and Development (DOST–PCIEERD)

The project is implemented under the Human Resource Development Program (HRDP) through the Visiting Expert Program (VEP) funding mechanism of DOST–PCIEERD.

The funding support is intended primarily for capacity-building, expert engagement, institutional learning, and interdisciplinary knowledge-transfer activities related to earthquake interpretation, historical seismic analysis, and methodological development within the scope of the approved VEP implementation.

D. Duration

Implementation Period: May 2026 – September 2026
Project Duration: Five (5) Months

The implementation timeline covers the approved schedule of lectures, mentoring sessions, interdisciplinary discussions, workshops, documentation activities, and synthesis sessions outlined in the Scheduled Plan of Activities and Session Design and Outline documents.

Adjustments to scheduling, modality, or sequencing of activities may be made when necessary due to institutional, logistical, or operational considerations, provided that such adjustments

remain consistent with the approved objectives, implementation framework, and reporting requirements of the Visiting Expert Program.

E. Project Leader

Ric L. Tura Jr.
Ateneo de Davao University Mathematics Department

The Project Leader shall be responsible for the day-to-day coordination and implementation of the program, including organizing activities, coordinating participants, overseeing documentation, communicating with the Visiting Expert and DOST–PCIEERD, and preparing institutional and project-related reports.

The Project Leader shall also ensure that all activities remain aligned with the approved objectives, scientific scope, implementation boundaries, and institutional responsibilities defined under these Terms of Reference and the Visiting Expert Program guidelines.

F. Visiting Expert

Dr. Jeffrey S. Perez
Supervising Science Research Specialist / Geologist
Philippine Institute of Volcanology and Seismology (DOST–PHIVOLCS)

The Visiting Expert shall serve as the primary source of domain expertise, methodological guidance, and professional insight for the program. Responsibilities include conducting expert-led lectures, facilitating interdisciplinary discussions, providing conceptual and methodological feedback, sharing professional practices relevant to earthquake interpretation and historical seismic analysis, and supporting institutional capacity-building efforts consistent with the objectives of the Visiting Expert Program.

The participation of the Visiting Expert shall be understood within the framework of expert engagement and knowledge transfer under the VEP implementation. It shall not be interpreted as constituting a formal institutional partnership, co-management arrangement, or operational endorsement beyond the approved scope of the program.

G. Collaborating Units

The project involves interdisciplinary participation and institutional coordination among the following units and participants:

- Ateneo de Davao University Mathematics Department
- Selected faculty members and student participants from Mathematics, Data Science, Computer Science, Computer Engineering, and related fields
- Participating trainees and project support personnel involved in documentation, coordination, and learning activities
- Visiting Expert engagement from the Philippine Institute of Volcanology and Seismology (DOST–PHIVOLCS)

The collaborating units and participants contribute to the program's interdisciplinary learning, analytical discussions, and knowledge-transfer activities within the approved capacity-building framework.

Participation in the program shall not automatically imply research authorship, intellectual ownership, formal research appointment, or inclusion in future publications or subsequent research initiatives beyond the scope of the Visiting Expert Program. Any future scholarly outputs, research activities, or collaborative undertakings arising after the implementation period shall be governed separately under appropriate institutional, ethical, and academic processes.

III. BACKGROUND AND RATIONALE

A. Background

The Philippines is situated within one of the most seismically active regions in the world, where complex tectonic interactions, active fault systems, subduction zones, and regional geodynamic processes shape earthquake hazards. Understanding the occurrence, characteristics, and impacts of earthquakes remains essential for scientific interpretation, disaster risk awareness, and long-term hazard understanding.

Historical and pre-instrumental earthquakes constitute an important component of Philippine seismicity. However, many of these events occurred before the establishment of modern instrumental monitoring systems, resulting in limited direct measurements and incomplete observational records. Consequently, the interpretation of such earthquakes frequently relies on macroseismic observations, historical accounts, intensity reports, geological context, and expert judgment.

Interpreting earthquake intensity observations poses both scientific and methodological challenges. Earthquake-related observations are often spatially uneven, uncertain, and incomplete. They are influenced by multiple interacting physical and environmental factors, including attenuation behavior, site effects, geological conditions, reporting limitations, and historical context. As a result, intensity-based earthquake interpretation requires careful reasoning that integrates seismological understanding, spatial interpretation, awareness of uncertainty, and interdisciplinary analytical perspectives.

Mathematical and computational approaches provide useful conceptual and analytical frameworks for organizing, interpreting, and examining earthquake-related information. However, the responsible application of such approaches requires a strong grounding in seismological principles, methodological limitations, observational uncertainty, and the realities of real-world earthquake interpretation. Quantitative reasoning alone is insufficient without an accompanying understanding of the physical, geological, and interpretative contexts in which earthquake observations are generated and analyzed.

In response to these considerations, the project is implemented under the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD through expert-guided interdisciplinary engagement focused on earthquake interpretation, historical seismic analysis, uncertainty-aware reasoning, and conceptual analytical workflows related to intensity-based earthquake epicenter estimation.

B. Rationale

The project recognizes the importance of strengthening institutional and individual competencies in seismology-informed earthquake interpretation, particularly within the context of historical and intensity-based seismic analysis. While existing mathematical, computational, and analytical methods may provide useful conceptual tools for studying earthquake observations, their responsible application requires deeper methodological understanding, interdisciplinary integration, and awareness of scientific limitations.

The Visiting Expert Program provides an opportunity for direct knowledge transfer, expert-guided mentoring, and interdisciplinary engagement involving professional practices in seismology, earthquake intensity interpretation, historical seismic analysis, and uncertainty assessment. Through expert-led discussions, consultations, workshops, and analytical sessions, the program seeks to strengthen conceptual understanding and methodological maturity among participating faculty members, trainees, and interdisciplinary participants.

The project also recognizes the growing importance of interdisciplinary approaches in modern earthquake-related studies. Contemporary seismic interpretation increasingly involves interactions among geoscientific reasoning, spatial analysis, uncertainty-aware interpretation, computational workflows, data organization, and exploratory analytical systems. As such, the program adopts an interdisciplinary structure involving participants from mathematics, data

science, computer science, computer engineering, and related fields to encourage broader scientific understanding and collaborative analytical thinking.

Rather than functioning as a formal software development or operational research implementation project under the Visiting Expert Program, the initiative focuses on building conceptual foundations, strengthening scientific reasoning, and developing research readiness through structured expert engagement. Discussions involving computational systems, geospatial interpretation, data-driven workflows, visualization environments, and emerging analytical approaches are intended to support interdisciplinary understanding and exploratory scientific learning within the scope of the approved VEP implementation.

The project further aims to strengthen institutional capacity within the host institution by supporting faculty development, interdisciplinary academic engagement, research-oriented learning, and long-term scholarly preparedness related to earthquake interpretation and historical seismicity. The knowledge, methodological insights, and interdisciplinary perspectives gained throughout the program are expected to support future instructional initiatives, research conceptualization, and academically grounded scientific engagement beyond the Visiting Expert Program.

C. Nature of the Program

The program is structured as an interdisciplinary capacity-building, knowledge-transfer, and research-readiness initiative implemented through the Visiting Expert Program mechanism of DOST–PCIEERD.

The implementation framework combines expert-led lectures, guided discussions, mentoring sessions, interdisciplinary analytical activities, conceptual workshops, documentation exercises, and synthesis sessions to strengthen scientific understanding and methodological awareness in earthquake interpretation and historical seismic analysis.

The program is intentionally designed to emphasize:

- uncertainty-aware scientific reasoning;
- interdisciplinary interpretation of earthquake observations;
- conceptual understanding of seismological and historical seismic analysis workflows;
- responsible interpretation of incomplete and spatially variable datasets;
- and methodological awareness in the use of mathematical, computational, and analytical approaches relevant to earthquake studies.

While selected sessions may include conceptual discussions involving computational workflows, geospatial systems, exploratory analytical methods, data-driven interpretation, visualization environments, artificial intelligence, machine learning, and prototype-oriented scientific discussions, such components are included solely within the context of exploratory learning, interdisciplinary exposure, and research-readiness development.

Accordingly, the program shall not be interpreted as:

- an operational earthquake prediction initiative;
- a validated forecasting or hazard deployment project;
- a formal software engineering implementation project;
- or a full-scale research and development undertaking under the Visiting Expert Program mechanism.

Any conceptual, analytical, or prototype-based outputs that may emerge during workshops, discussions, or synthesis activities shall be treated as exploratory educational outputs intended solely for learning, scientific reflection, interdisciplinary integration, and future academic planning.

The program further recognizes that participation in interdisciplinary analytical and exploratory activities does not automatically establish research authorship, intellectual ownership, formal research appointments, or institutional partnership arrangements beyond the approved scope of the Visiting Expert Program and existing institutional policies.

All project activities, participant engagements, documentation procedures, and exploratory outputs shall remain subject to the scientific, administrative, ethical, and institutional guidelines established under these Terms of Reference, the approved project framework, and applicable DOST–PCIEERD policies.

IV. OBJECTIVES

A. General Objective

To strengthen competencies in seismology-informed, intensity-based earthquake epicenter estimation via direct knowledge transfer from a PHIVOLCS Visiting Expert.

B. Specific Objectives

1. To deepen understanding of earthquake intensity scales, interpretation practices, and historical limitations.
2. To learn expert methods for validating epicenter estimates and uncertainty assessment.
3. To refine conceptual and methodological frameworks for intensity-based analysis through mentoring.
4. To document insights and best practices to inform future research and instruction.

C. Scope and Methodological Limitations

The objectives of the program shall be interpreted within the framework of the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD, which primarily supports expert engagement, knowledge transfer, institutional learning, and interdisciplinary capacity-building activities.

Accordingly, the project is intended to strengthen conceptual understanding, methodological awareness, scientific reasoning, and interdisciplinary analytical preparedness related to earthquake interpretation, historical seismic analysis, and intensity-based earthquake epicenter estimation.

The project does not seek to produce operational earthquake prediction systems, validated forecasting technologies, deployable hazard platforms, or full-scale research and development implementations under the Visiting Expert Program mechanism.

While the program may include conceptual and exploratory discussions involving:

1. computational workflows;
2. geospatial systems;
3. visualization environments;
4. uncertainty-aware analytical approaches;
5. data-driven interpretation;
6. machine learning concepts;
7. artificial intelligence applications;
8. and prototype-oriented scientific exercises, such activities shall be understood strictly within the context of:
    - educational exploration;
    - interdisciplinary exposure;
    - conceptual analysis;
    - research-readiness development;
    - and methodological understanding.

Any conceptual, analytical, or prototype-based outputs produced during workshops, synthesis activities, or interdisciplinary sessions shall not be interpreted as operational scientific products, validated seismic technologies, institutional hazard tools, or formal research deliverables under the approved VEP implementation.

The project further recognizes that earthquake interpretation inherently involves uncertainty, observational limitations, incomplete datasets, and context-dependent scientific reasoning. Consequently, discussions and activities conducted under the program shall emphasize responsible scientific interpretation, methodological caution, and uncertainty-aware analytical thinking rather than deterministic or predictive claims.

All project activities shall remain aligned with:

- the approved objectives of the Visiting Expert Program;
- the provisions of the Memorandum of Agreement (MOA);
- the Session Design and Outline;
- the Scheduled Plan of Activities;
- and applicable institutional and DOST–PCIEERD policies and guidelines.

Any future research activities, software implementations, publications, or extended interdisciplinary initiatives that may emerge after the completion of the program shall be pursued separately, with appropriate institutional review, ethical processes, authorship agreements, and funding mechanisms beyond the scope of the present Visiting Expert Program.

V. SCOPE OF WORK AND RESPONSIBILITIES

A. Visiting Expert

The Visiting Expert shall serve as the primary source of technical, methodological, and professional expertise for the implementation of the Visiting Expert Program.

The Visiting Expert shall provide expert-guided engagement related to:

- seismology-informed earthquake interpretation;
- historical and macroseismic analysis;
- earthquake intensity interpretation practices;
- uncertainty-aware reasoning;
- tectonic and geological context;
- methodological considerations in intensity-based earthquake epicenter estimation;
- and interdisciplinary analytical perspectives relevant to the objectives of the program.
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The responsibilities of the Visiting Expert include, but are not limited to, the following:

1. Conduct expert-led lectures, seminars, discussions, and guided learning sessions aligned with the approved Session Design and Outline and Scheduled Plan of Activities;
2. Facilitate conceptual and interdisciplinary discussions involving earthquake interpretation, historical seismic analysis, uncertainty assessment, and methodological reasoning;
3. Provide professional insight and mentoring based on practical experience in seismology, earthquake hazard interpretation, and related scientific workflows;
4. Guide participants in understanding the strengths, limitations, assumptions, and appropriate interpretation of earthquake-related observational data and analytical approaches;
5. Participate in synthesis sessions, conceptual reviews, and interdisciplinary discussions involving exploratory analytical or prototype-oriented outputs developed during the program;
6. Provide methodological feedback and conceptual guidance related to scientific interpretation, analytical reasoning, and interdisciplinary understanding within the approved scope of the Visiting Expert Program;

7. Support institutional capacity-building and knowledge-transfer efforts consistent with the objectives of DOST–PCIEERD HRDP–VEP.

The role of the Visiting Expert shall remain within the framework of expert engagement, mentoring, and knowledge transfer under the approved VEP implementation. Participation of the Visiting Expert in lectures, consultations, discussions, reviews, or exploratory activities shall not, in itself, constitute co-development, institutional endorsement, a formal research partnership, or authorship of future outputs beyond any separately established agreements and applicable institutional policies.

B. Project Team

The Project Team shall be responsible for the overall coordination, implementation, support, documentation, participant management, and administrative execution of the program.

The Project Team shall operate under the supervision of the Project Leader and in coordination with the Visiting Expert, the implementing institution, and DOST–PCIEERD requirements.

Responsibilities of the Project Team include, but are not limited to, the following:

1. Coordinate the scheduling, organization, and implementation of all activities included in the approved Session Design and Outline and Scheduled Plan of Activities;
2. Facilitate communication among the Visiting Expert, participants, implementing institution, and relevant coordinating offices;
3. Ensure proper documentation of lectures, mentoring sessions, discussions, attendance, outputs, and implementation activities in accordance with institutional and DOST–PCIEERD reporting requirements;
4. Prepare and consolidate session summaries, reflection outputs, activity documentation, and other learning-related materials generated during the implementation of the program;
5. Assist in logistical arrangements involving online and onsite sessions, participant coordination, venue preparation, and communication workflows;
6. Ensure that all project activities remain aligned with:
    - the approved objectives of the Visiting Expert Program;
    - the provisions of the Memorandum of Agreement (MOA);
    - the approved implementation framework;
    - and applicable institutional and DOST–PCIEERD policies and guidelines;
7. Facilitate an environment of responsible scientific discussion, interdisciplinary collaboration, and uncertainty-aware reasoning consistent with the educational and methodological objectives of the program;
8. Support monitoring, evaluation, and preparation of institutional and project-related reports required under the Visiting Expert Program implementation.

The Project Team shall also ensure that exploratory discussions, conceptual analytical activities, and any prototype-oriented exercises conducted during the program remain educational and research-readiness-oriented and are not represented as operational scientific systems or validated earthquake technologies under the scope of the VEP implementation.

C. Participant Structure and Involvement

The program adopts an interdisciplinary participant structure designed to support knowledge transfer, methodological understanding, scientific exposure, and collaborative analytical learning across multiple academic and technical perspectives.

Participants may include:

- faculty members;
- student trainees;
- project support personnel;

- and selected interdisciplinary participants from fields such as mathematics, data science, computer science, computer engineering, earth science, and related disciplines.

Participant involvement in the program is intended primarily for:

- learning and capacity-building;
- interdisciplinary engagement;
- methodological understanding;
- conceptual exploration;
- research-readiness development;
- and participation in guided scientific discussions and structured analytical activities.

Participants may engage in:

- lectures and expert-led discussions;
- conceptual workshops;
- case-study analyses;
- interdisciplinary discussions;
- documentation activities;
- uncertainty-aware interpretation exercises;
- exploratory analytical activities;
- and conceptual or prototype-oriented synthesis sessions consistent with the approved implementation framework.

Participation in the program does not automatically imply:

- research authorship;
- intellectual ownership;
- formal research appointments;
- institutional representation;
- ownership of conceptual outputs;
- or inclusion in future publications, research initiatives, software development efforts, or extended collaborations beyond separately established agreements and applicable institutional processes.

Any conceptual, analytical, or prototype-based outputs developed during the program shall be treated solely for:

- interdisciplinary learning;
- scientific discussion;
- conceptual synthesis;
- and research-readiness purposes within the approved scope of the Visiting Expert Program.

Participants are expected to:

- observe professional and ethical conduct during all program activities;
- respect institutional and scientific guidelines;
- participate responsibly in discussions and learning activities;
- and contribute to a collaborative and academically grounded learning environment.

The project further recognizes that interdisciplinary engagement may involve differing technical backgrounds, disciplinary perspectives, and levels of expertise. Accordingly, activities shall emphasize conceptual clarity, responsible interpretation, collaborative learning, and methodological awareness rather than technical specialization alone.

All participant activities and engagements shall remain subject to the implementation boundaries, scientific limitations, institutional guidelines, and governance provisions established under these Terms of Reference and applicable DOST–PCIEERD policies.

VI. IMPLEMENTATION PLAN

A. Workplan

The implementation of the Visiting Expert Program shall follow a phased interdisciplinary capacity-building framework designed to support structured knowledge transfer, methodological understanding, uncertainty-aware reasoning, and exploratory scientific engagement related to earthquake interpretation and intensity-based seismic analysis.

(1) OBJECTIVES *(with % weight)*	**(2) TARGET ACTIVITIES**	**(3) TARGET ACCOMPLISHMENT** *(with % weight and objectively verifiable indicators)*	**(4) MONTHLY TARGETS**					
			Y1 ***(100%)*** *(with % weight per month)*					
			May	**June**	**July**	**August**	**September**	**Total**
1. Human Resource Development Program (Foundational and Conceptual Understanding) – (40%) To strengthen foundational understanding in seismology-informed earthquake interpretation, observational uncertainty, intensity-based reasoning, and interdisciplinary analytical perspectives relevant to historical and contemporary seismic events.	Expert-led lectures and interdisciplinary discussions on: 1. Earthquake processes, seismic observations, macroseismic interpretation, attenuation behavior, and uncertainty-aware scientific reasoning. 2. Guided discussions on earthquake observations, spatial interpretation, and tectonic contexts in the Philippine setting. 3. Conceptual discussions on modern seismological workflows and data-driven scientific interpretation.	**Compiled Learning Portfolio and Reflection Outputs (15%)** Participants are expected to develop a foundational conceptual understanding of earthquake interpretation, observational systems, and uncertainty-aware reasoning through interdisciplinary engagement and guided reflection activities. **Session Documentation and Conceptual Exercises (15%)** Participants are expected to complete conceptual analytical exercises, interdisciplinary discussions, and documented learning activities aligned with the session framework. **Interdisciplinary Participation and Engagement (10%)** Participants are expected to engage in lectures, workshops, discussions actively, and guided analytical sessions across interdisciplinary learning environments.	15%	15%	5%	3%	2%	40%
2. Human Resource Development Program (Methodological Understanding and Scientific Interpretation) – (35%) To strengthen methodological awareness, interdisciplinary scientific reasoning, and conceptual understanding of historical seismic interpretation, uncertainty assessment, and data-informed earthquake analysis through expert-guided mentoring and case-study discussions.	1. Guided discussions and mentoring sessions involving historical earthquake catalogs, macroseismic interpretation, observational uncertainty, and conceptual earthquake inference. 2. Interdisciplinary case-study analyses involving selected Philippine earthquakes and observational	**Historical and Case Study Analysis Outputs (15%)** Participants are expected to demonstrate understanding of historical earthquake interpretation, observational limitations, and uncertainty-aware scientific reasoning through documented analytical outputs and guided case-study activities. **Conceptual Reflection and Methodological Synthesis (10%)** Participants are expected to		5%	15%	10%	5%	35%

	3. complexity. Conceptual review sessions involving uncertainty-aware interpretation, exploratory analytical workflows, and interdisciplinary scientific reasoning.	synthesize methodological insights and interdisciplinary understanding gained from mentoring sessions and guided analytical discussions. **Documentation of Scientific and Methodological Insights (10%)** Participants are expected to contribute to session summaries, conceptual notes, and documentation materials supporting institutional learning and future academic engagement.						
3. Human Resource Development Program (Interdisciplinary Integration and Research Readiness) – (25%) To strengthen interdisciplinary scientific integration, exploratory analytical understanding, and research-readiness through guided discussions on modern seismic workflows, conceptual systems analysis, and synthesis-oriented activities.	1. Interdisciplinary discussions involving computational workflows, geospatial interpretation, visualization systems, and exploratory analytical approaches in seismology. 2. Guided synthesis activities and conceptual workshops involving interdisciplinary scientific integration. 3. Prototype-oriented presentation and scientific review sessions for exploratory educational outputs developed during the program.	**Conceptual System and Analytical Outputs (10%)** Participants are expected to demonstrate interdisciplinary understanding of conceptual scientific systems, exploratory analytical approaches, and uncertainty-aware workflows relevant to earthquake interpretation. **Prototype or Conceptual Presentation Outputs (10%)** Participants are expected to present exploratory conceptual or prototype-based outputs intended solely for educational, interdisciplinary, and research-readiness purposes. **Certificate of Completion and Final Integration Activities (5%)** Participants who complete the required sessions, activities, and synthesis outputs shall receive recognition of participation and completion under the Visiting Expert Program framework.	0%	0%	0%	10%	15%	25%

The work plan integrates expert-led discussions, mentoring activities, interdisciplinary analytical sessions, conceptual workshops, reflection exercises, and synthesis activities consistent with the approved Session Design and Outline and Scheduled Plan of Activities.

All activities shall remain aligned with the approved objectives of the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP), the Memorandum of Agreement (MOA), and applicable DOST–PCIEERD guidelines and institutional policies.

The implementation framework is intended primarily for educational, methodological, interdisciplinary, and research-readiness purposes. Any conceptual, analytical, or prototype-oriented activities conducted during the program shall remain exploratory and non-operational.

The monthly targets and percentage allocations reflected in the work plan are intended to guide the sequencing and implementation flow of the program. Minor adjustments to activity timing,

delivery modality, or scheduling may be implemented when necessary due to operational, institutional, or logistical considerations, provided that such adjustments remain consistent with the approved objectives, implementation framework, and reporting requirements of the Visiting Expert Program.

All implementation activities shall remain subject to the implementation boundaries, methodological limitations, participant governance provisions, scientific considerations, and institutional guidelines established under these Terms of Reference, the Memorandum of Agreement (MOA), the Session Design and Outline, the Scheduled Plan of Activities, and applicable DOST–PCIEERD policies.

B. Scheduled Plan of Activities

The Scheduled Plan of Activities provides the operational implementation schedule of the Visiting Expert Program. It serves as the primary reference for the sequencing, coordination, and conduct of lectures, mentoring sessions, interdisciplinary discussions, workshops, documentation activities, and synthesis sessions.

Month	Activity	Description of Activities	Expected Outputs / Deliverables	Responsible Unit / Person
May 2026	Foundations of Earthquake Processes, Observations, and Hazard Interpretation	Conduct introductory interdisciplinary sessions discussing the physical foundations of earthquakes, seismic waves, earthquake observations, intensity interpretation, and hazard-related concepts. Discussions will emphasize the distinction between earthquake source mechanisms, wave propagation, observational data, and damage interpretation. Participants will be introduced to earthquakes as measurable, uncertain phenomena relevant to mathematical, computational, and geoscientific analysis.	Compiled Learning Portfolio; Session Documentation; Participant Reflection Outputs; Foundational Concept Notes	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees
June 2026	Tectonic Sources and Spatial Interpretation of Philippine Earthquakes	Conduct guided discussions on tectonic structures and earthquake-generating systems in the Philippine setting. Activities include conceptual interpretation of earthquake locations, spatial variability of seismic intensity, attenuation behavior, and geological controls affecting observations. Emphasis will be placed on how spatial reasoning and observational interpretation contribute to earthquake analysis workflows.	Compiled Learning Portfolio; Spatial Interpretation Exercises; Technical Discussion Outputs; Attendance and Documentation Reports	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees
July 2026	Historical Earthquake Catalogs, Macroseismic Data, and Observational Uncertainty	Facilitate sessions examining historical earthquake records, macroseismic observations, and uncertainty in earthquake catalogs. Discussions will focus on limitations of historical data, reporting bias, incomplete observations, and interpretative challenges in reconstructing earthquake events from non-instrumental records. Participants will examine the role of uncertainty in	Compiled Learning Portfolio; Historical Data Interpretation Exercises; Reflection and Analysis Outputs; Session Documentation	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees

		scientific inference and historical seismic interpretation.		
August 2026	Case Studies of Significant Philippine Earthquakes: Interpretation, Uncertainty, and Data-Driven Analysis	Conduct interdisciplinary case study discussions on significant Philippine earthquakes relevant to the expertise of the Visiting Expert. Sessions will analyze observational patterns, seismic intensity distributions, interpretation uncertainty, and implications for data-driven analysis and hazard understanding. Participants will examine how real earthquake events are studied through scientific and computational workflows.	Compiled Learning Portfolio; Case Study Analysis Outputs; Technical Notes and Discussions; Activity Documentation	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees
September 2026	Modern Seismological Workflows, Data Systems, and Challenges in the Philippine Context	Conduct discussions on modern seismological workflows, earthquake data systems, mapping platforms, computational tools, and scientific data management approaches relevant to seismic analysis. Emphasis will be placed on challenges in the Philippine setting, including uneven instrumentation, historical data gaps, observational uncertainty, and interdisciplinary opportunities for mathematical, computational, and data-driven system development. Participants will also be introduced to conceptual considerations relevant to the development of scientific software systems for earthquake analysis.	Compiled Learning Portfolio; Conceptual System Design Outputs; Technical Reflection Papers; Session Documentation and Summary Report	Visiting Expert; Project Leader; Mathematics Department; Project Team; Participating Trainees

C. Timeline

The schedule is designed to ensure progressive interdisciplinary learning and conceptual integration across the program's five major phases. Activities are structured to gradually develop participant understanding in earthquake interpretation, tectonic and historical seismic analysis, uncertainty-aware reasoning, interdisciplinary scientific workflows, and exploratory analytical approaches relevant to intensity-based earthquake epicenter estimation.

The implementation schedule shall remain consistent with:

- the approved objectives of the Visiting Expert Program;
- the Memorandum of Agreement (MOA);
- the Session Design and Outline;
- the approved Workplan;
- and applicable institutional and DOST–PCIEERD implementation guidelines.

Minor adjustments to scheduling, activity sequencing, or delivery modality may be implemented when necessary due to operational, institutional, logistical, or force majeure considerations, provided that such adjustments do not materially alter the approved objectives, implementation framework, or educational direction of the program.

D. Risk Management Plan

The Risk Management Plan identifies potential operational, scientific, administrative, institutional, and implementation-related risks that may affect the conduct of the Visiting Expert Program under the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD.

(12) Risk Management Plan								
Objective	**Risks**	**Risk Rating**						**(13) Mitigation/Action Plan (Use separate sheet if necessary)**
		Impacts on Timeline		**Impact on Budget**		**Probability**		
		Rating	**Reason**	**Rating**	**Reason**	**Rating**	**Reason**	
1. To deepen understanding of earthquake intensity scales, interpretation practices, and historical limitations.	Delays in scheduling or implementation of expert-led sessions due to institutional, operational, or scheduling constraints.	Moderate	Delays in lectures or mentoring sessions may affect the sequencing of foundational learning activities and scheduled outputs.	Low	Most activities are conducted online and involve minimal additional operational costs.	Moderate	Scheduling coordination among institutions, participants, and the Visiting Expert may require adjustments.	Maintain flexible scheduling arrangements; coordinate schedules in advance; prepare alternative online delivery mechanisms and asynchronous learning materials when necessary.
1. To deepen understanding of earthquake intensity scales, interpretation practices, and historical limitations.	Participant disengagement or uneven technical background among trainees.	Moderate	Variations in participant preparedness may slow interdisciplinary discussions and conceptual progression.	Low	Does not significantly affect the approved implementation budget.	Moderate	Participants come from multiple disciplines with differing technical backgrounds and levels of familiarity with seismology.	Conduct orientation and leveling discussions; emphasize conceptual clarity; provide supplementary reading materials and guided discussions to support interdisciplinary understanding.
2. To learn expert methods for validating epicenter estimates and uncertainty assessment.	Difficulty in interpreting incomplete, uncertain, or historically inconsistent earthquake data.	Moderate	May affect the pace of analytical discussions and uncertainty-aware exercises.	Low	Primarily affects learning activities rather than requiring additional expenditures.	High	Historical and macroseismic datasets inherently contain observational limitations and interpretative ambiguity.	Emphasize uncertainty-aware reasoning and methodological limitations during discussions; incorporate guided case-study analysis and conceptual interpretation exercises.
2. To learn expert methods for validating epicenter estimates and uncertainty assessment.	Limited availability of historical or reference materials relevant to selected case studies.	Moderate	May require adjustments to planned case-study activities or supplementary materials.	Low	Additional resources may be sourced digitally without significant budget impact.	Moderate	Some historical earthquake records and reference materials may be incomplete or difficult to access.	Prepare alternative case studies and supplementary references; coordinate with the Visiting Expert regarding appropriate substitute materials and examples.
3. To refine	Misinterpretation of	High	Misinterpretation may	Moderate	Clarifications,	Moderate	The project includes	Consistently reinforce

conceptual and methodological frameworks for intensity-based analysis through mentoring.	exploratory analytical outputs as operational scientific systems or predictive technologies.		affect implementation direction, documentation, and institutional alignment.		revisions, or institutional coordination may become necessary if outputs are publicly misunderstood.		exploratory discussions involving computational systems and conceptual prototypes that may be misunderstood outside the approved scope.	implementation boundaries, methodological limitations, and exploratory framing throughout lectures, reports, documentation, and presentations.
3. To refine conceptual and methodological frameworks for intensity-based analysis through mentoring.	Scope expansion beyond approved VEP objectives due to emerging interdisciplinary interests or prototype-oriented discussions.	Moderate	Uncontrolled scope expansion may affect implementation pacing and completion of scheduled activities.	Moderate	Additional activities outside approved scope may require unsupported expenditures or administrative adjustments.	Moderate	Interdisciplinary engagement may naturally generate additional project ideas beyond the approved implementation framework.	Ensure all activities remain aligned with the approved ToR, MOA, Workplan, and Scheduled Plan of Activities; subject proposed extensions to separate institutional review processes if necessary.
4. To document insights and best practices to inform future research and instruction.	Delays in documentation consolidation, report preparation, or submission of implementation outputs.	Moderate	Delays may affect reporting schedules and final consolidation of project outputs.	Low	Primarily administrative in nature and unlikely to require major additional expenditures.	Moderate	Documentation activities depend on coordination among project personnel, participants, and implementation schedules.	Assign documentation focal persons early; establish internal deadlines and standardized documentation procedures; conduct periodic progress reviews.
4. To document insights and best practices to inform future research and instruction.	Ambiguity regarding participant roles, authorship expectations, or future scholarly outputs.	Moderate	Misunderstandings may affect collaboration and project coordination.	Low	Does not significantly affect the implementation budget.	Moderate	The interdisciplinary and exploratory nature of the program may create differing expectations among participants.	Clearly communicate participant roles, authorship limitations, and implementation boundaries through orientations, documentation, and the Terms of Reference.
General Project Implementation	Budgetary limitations, delayed fund processing, or administrative processing delays.	Moderate	Delays in fund processing may affect scheduling, logistics, and implementation coordination.	High	Delayed releases may temporarily affect implementation support and	Moderate	Government and institutional processing timelines may vary.	Maintain regular coordination with DOST–PCIEERD and institutional offices; prioritize essential implementation activities;

					operational coordination.			maximize online implementation and resource-sharing approaches when necessary.
General Project Implementation	Technical or connectivity issues during online implementation activities.	Low	Temporary interruptions may affect online sessions or participant engagement.	Low	Minimal additional costs expected for contingency arrangements.	Moderate	Online implementation depends on internet connectivity and digital platform accessibility.	Prepare backup communication platforms, session recordings, and asynchronous materials; coordinate technical support before major sessions.

The plan is intended to support proactive monitoring, responsible implementation, and continuity of project activities by identifying possible risks, evaluating their potential effects on the project timeline and budget, and outlining corresponding mitigation and response strategies.

The identified risks are assessed according to their possible impact on:

- the timely achievement of project objectives;
- the implementation schedule and deliverables;
- the approved project budget and operational resources;
- and the overall conduct of interdisciplinary capacity-building and knowledge-transfer activities under the approved framework of the program.

Risk ratings are categorized as High, Moderate, or Low based on the estimated severity of impact and likelihood of occurrence. The mitigation strategies identified under this section are intended to reduce implementation disruptions, maintain institutional and scientific consistency, and ensure that all project activities remain aligned with the approved objectives, implementation boundaries, and governance provisions established under the Terms of Reference, the Memorandum of Agreement (MOA), and applicable DOST–PCIEERD and institutional guidelines.

VII. EXPECTED OUTPUTS

The expected outputs of the Visiting Expert Program are intended primarily to support interdisciplinary capacity-building, methodological strengthening, institutional learning, and research-readiness development consistent with the objectives of the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD.

(10) EXPECTED OUTPUTS (6Ps)	YEARLY AND QUARTERLY TARGETS						Grand Total
	Y1						
	May	Jun	Jul	Aug	Sep	Total	
Publications	Identification of possible scholarly directions, literature consolidation, and development of methodological notes related to earthquake interpretation and historical seismic analysis.	Preparation of technical summaries and interdisciplinary discussion syntheses relevant to tectonic interpretation and observational analysis.	Consolidation of analytical reflections, historical seismic interpretation notes, and uncertainty-aware conceptual documentation.	Preparation of interdisciplinary synthesis materials and exploratory analytical summaries derived from case-study discussions and guided workshops.	Compilation of institutional reference materials and exploratory scholarly outputs that may support future publication initiatives subject to separate institutional review and research development processes.	**1**	**1**
Patents/IP	-	-	-	-	-	**0**	**0**
Products	-	-	-	-	-	**0**	**0**
People Services	Conduct of foundational interdisciplinary training sessions on earthquake processes, seismic observations, and hazard interpretation involving approximately thirty (30) non-formally trained participants from mathematics, data science, computer science, computer engineering, earth science, and related fields.	Conduct of expert-led discussions and guided learning activities on tectonic sources, spatial interpretation, attenuation behavior, and earthquake-generating systems in the Philippine setting for approximately thirty (30) participants.	Conduct of workshops and interdisciplinary discussions on historical earthquake catalogs, macroseismic interpretation, uncertainty-aware reasoning, and observational limitations involving approximately thirty (30) participants.	Conduct of interdisciplinary case-study discussions and guided analytical activities involving significant Philippine earthquakes, observational complexity, and conceptual data-driven analysis for approximately thirty (30) participants.	Conduct of onsite synthesis sessions, conceptual workshops, exploratory scientific systems discussions, and final integration activities involving approximately thirty (30) participants; selected project team members further capacitated through direct involvement in project coordination, synthesis activities, documentation, and exploratory prototype development.	**31**	**31**
Places and Partnerships	Institutional coordination and interdisciplinary engagement activities were initiated within the Ateneo de Davao University Mathematics Department and participating academic units under the approved implementation framework.	Strengthening of interdisciplinary academic collaboration involving mathematics, data science, computer science, computer engineering, earth science, and related fields through collaborative learning and scientific engagement activities.	Continued expert-guided academic engagement and institutional coordination supporting historical seismic interpretation and uncertainty-aware scientific discussions.	Exploration of possible future academic, scientific, or institutional collaborations related to earthquake interpretation, historical seismic analysis, and interdisciplinary scientific engagement, subject to appropriate institutional review and approval processes.	Consolidation of institutional learning networks, interdisciplinary academic engagement, and exploratory collaborative interactions developed during the Visiting Expert Program implementation.	**1**	**1**
Policy	-	-	-	-	-	**0**	**0**

Total Yearly Target (6Ps): 34

Notes:

1. All outputs identified under this table shall be interpreted within the framework of interdisciplinary capacity-building, institutional learning, methodological development, and research-readiness enhancement under the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD.
2. The “People Services” component distinguishes between:
    - participants trained through non-formal interdisciplinary training activities under the Visiting Expert Program; and
    - project team members further capacitated through direct involvement in implementation, coordination, documentation, synthesis activities, and exploratory analytical engagements related to the project.
3. Any conceptual, analytical, or prototype-oriented outputs generated during the implementation period shall remain exploratory and educational in nature and shall not be interpreted as operational scientific systems, validated hazard technologies, or formal research and development products under the Visiting Expert Program implementation.
4. Any future scholarly publications, institutional partnerships, software initiatives, interdisciplinary projects, or extended research activities that may arise after the implementation period shall be pursued separately under appropriate institutional review, authorship agreements, ethical processes, and applicable funding or implementation mechanisms beyond the scope of the current Visiting Expert Program.

The outputs identified under this section shall be interpreted within the framework of:

- expert-guided knowledge transfer;
- interdisciplinary scientific engagement;
- uncertainty-aware analytical reasoning;
- conceptual and methodological development;
- and exploratory educational activities related to earthquake interpretation and historical seismic analysis.

Accordingly, the expected outputs shall not be interpreted as constituting:

- operational earthquake prediction systems;
- validated hazard technologies;
- deployable seismic platforms;
- or formal research and development deliverables under the Visiting Expert Program implementation.

All outputs generated during the implementation period shall remain subject to:

- the approved scope and methodological limitations of the program;
- institutional and DOST–PCIEERD policies;
- participant governance provisions;
- and applicable scientific, ethical, and documentation guidelines established under these Terms of Reference.

A. Technical Outputs

The program's technical outputs shall focus primarily on conceptual understanding, interdisciplinary analytical engagement, scientific documentation, and outputs generated through the approved implementation framework.

Expected technical outputs may include:

1. Compiled Learning Portfolios containing:
    - participant reflection outputs;
    - conceptual notes;
    - session summaries;
    - analytical exercises;
    - interdisciplinary discussion outputs;
    - and synthesized methodological insights derived from lectures, workshops, mentoring sessions, and guided discussions.

2. Session Documentation and Activity Records involving:
    - attendance documentation;
    - technical discussion summaries;
    - mentoring session documentation;
    - activity records;
    - and interdisciplinary analytical outputs generated during the implementation period.

3. Conceptual and Analytical Outputs related to:
    - uncertainty-aware interpretation;
    - historical earthquake analysis;
    - spatial interpretation exercises;
    - exploratory analytical reasoning;
    - geospatial interpretation;
    - and conceptual scientific workflow discussions relevant to earthquake interpretation and intensity-based seismic analysis.

4. Exploratory Conceptual or Prototype-Oriented Educational Outputs developed during synthesis sessions, interdisciplinary discussions, workshops, or guided analytical activities for educational and research-readiness purposes only.

Any conceptual, analytical, or prototype-oriented outputs generated under the program shall be treated solely as outputs and shall not be interpreted as:

- operational software systems;
- validated analytical platforms;
- deployable seismic technologies;
- institutional hazard systems;
- or formal research and development products under the Visiting Expert Program.

The technical outputs generated during the implementation period are intended primarily to support:

- interdisciplinary scientific understanding;
- methodological reflection;
- institutional learning;
- conceptual integration;
- and future academic planning beyond the scope of the current VEP implementation.

B. Capacity Building Outputs

The primary outputs of the program are expected to be realized through strengthened human and institutional capacity in seismology-informed earthquake interpretation, historical seismic analysis, uncertainty-aware reasoning, and interdisciplinary scientific engagement.

Expected capacity-building outputs include:

1. Enhanced conceptual understanding among participating faculty members, trainees, and interdisciplinary participants regarding:
    - earthquake intensity interpretation;
    - historical earthquake analysis;
    - tectonic and spatial reasoning;
    - uncertainty-aware scientific interpretation;
    - and interdisciplinary analytical workflows.

2. Strengthened interdisciplinary engagement involving participants from mathematics, data science, computer science, computer engineering, earth science, and related fields through collaborative discussions, guided analytical activities, and conceptual workshops.

3. Increased methodological awareness and research-readiness among participants through:
    - expert-guided mentoring;
    - conceptual synthesis activities;
    - analytical reflection exercises;
    - and interdisciplinary scientific discussions.
4. Improved awareness of:
    - observational uncertainty;
    - scientific limitations;
    - responsible interpretation of earthquake-related information;
    - and the importance of interdisciplinary reasoning in seismic interpretation and historical earthquake analysis.

5. Development of institutional learning resources through the consolidation of:
    - methodological notes;
    - synthesized discussion outputs;
    - conceptual analytical materials;
    - and session-based documentation supporting future instructional and academic activities.

The program's capacity-building outputs are intended to contribute to the long-term strengthening of academic competencies, methodological maturity, and interdisciplinary scientific engagement within the host institution and participating academic communities.

Participation in capacity-building activities under the program shall not automatically establish:

- formal research appointments;
- authorship rights;
- intellectual ownership;
- institutional representation;
- or guaranteed inclusion in future publications, software development efforts, or extended collaborative initiatives beyond separately established agreements and institutional processes.

C. Institutional Outputs

The program is expected to contribute to institutional strengthening through the development of structured interdisciplinary learning activities, collaborative scientific engagement, documentation systems, and enhanced methodological understanding related to earthquake interpretation and historical seismic analysis.

Expected institutional outputs include:

1. Consolidated implementation documentation involving:
    - session records;
    - activity reports;
    - reflection outputs;
    - mentoring documentation;

- and interdisciplinary analytical summaries supporting institutional learning and reporting requirements.

2. Strengthened institutional experience in implementing interdisciplinary, uncertainty-aware, and scientifically grounded capacity-building activities involving earthquake interpretation and historical seismic analysis.

3. Enhanced academic and professional engagement among participating units, faculty members, trainees, and expert collaborators involved in the implementation of the Visiting Expert Program.

4. Development of institutional reference materials and methodological learning resources that may support:
   - future instructional activities;
   - academic mentoring;
   - interdisciplinary learning;
   - and subsequent research conceptualization beyond the scope of the current VEP implementation.

5. Strengthened institutional preparedness for future scholarly, interdisciplinary, or research-oriented initiatives related to:
   - seismology-informed earthquake interpretation;
   - historical seismicity;
   - uncertainty-aware analysis;
   - and exploratory scientific systems discussions.

The program's institutional outputs are intended primarily for academic, educational, methodological, and institutional development.

Any future research activities, publications, software initiatives, formal interdisciplinary projects, or extended collaborations that may arise after the implementation period shall be pursued separately, with appropriate institutional review, authorship agreements, ethical processes, and applicable funding mechanisms, beyond the scope of the current Visiting Expert Program implementation.

All expected outputs identified under this section shall remain subject to the implementation limitations, scientific boundaries, governance provisions, and institutional guidelines established under these Terms of Reference and applicable DOST–PCIEERD policies.

VIII. DELIVERABLES AND REPORTING

The deliverables and reporting requirements of the Visiting Expert Program are intended to support:

- implementation monitoring;
- institutional accountability;
- documentation of interdisciplinary learning activities;
- and compliance with the requirements of DOST–PCIEERD and the approved implementation framework.

All reports, documentation materials, and deliverables generated under the program shall remain aligned with:

- the approved objectives of the Visiting Expert Program;
- the Memorandum of Agreement (MOA);
- the approved Session Design and Outline;
- the Scheduled Plan of Activities;
- the approved Workplan;
- and applicable institutional and DOST–PCIEERD policies and guidelines.

The deliverables identified under this section are intended primarily to document:

- capacity-building activities;

- interdisciplinary scientific engagement;
- methodological development;
- implementation progress;
- and institutional learning outcomes.

Accordingly, the required deliverables shall not be interpreted as constituting:

- formal research and development outputs;
- operational software deployment reports;
- validated hazard systems;
- or technical certification of analytical or prototype-oriented educational outputs developed during the implementation period.

All documentation generated during the program shall remain subject to:

- institutional reporting requirements;
- ethical and scientific considerations;
- participant governance provisions;
- and applicable confidentiality, authorship, and documentation policies.

A. Narrative Reports

Narrative reports shall be prepared to document progress, major activities conducted, interdisciplinary learning engagements, mentoring sessions, conceptual workshops, and significant observations arising during the program's implementation.

Narrative reports may include:

- summaries of lectures and mentoring sessions;
- implementation updates;
- interdisciplinary discussion highlights;
- synthesis of learning outcomes;
- participant engagement observations;
- and reflections on methodological and institutional development achieved during the implementation period.

The Project Team shall ensure that narrative reports:

- accurately reflect the approved scope and objectives of the program;
- maintain scientific and institutional consistency;
- and avoid unsupported claims regarding operational systems, predictive capabilities, or validated technologies beyond the approved scope of the Visiting Expert Program.

Narrative reports shall be submitted in accordance with applicable DOST–PCIEERD reporting schedules and institutional requirements.

B. Attendance

Attendance records shall be maintained for all lectures, workshops, mentoring sessions, interdisciplinary discussions, synthesis activities, and related implementation activities conducted under the Visiting Expert Program.

Attendance documentation may include:

- participant attendance sheets;
- session participation records;
- online meeting records;
- onsite participation logs;
- and other documentation necessary for institutional and DOST–PCIEERD monitoring purposes.

The Project Team shall ensure the proper consolidation, safekeeping, and submission of attendance records as part of the program documentation and reporting process.

Attendance records shall be used solely for:

- implementation monitoring;
- participant engagement tracking;
- institutional documentation;
- and certification purposes within the approved scope of the program.

Participation in program activities and attendance documentation shall not automatically establish:

- authorship rights;
- intellectual ownership;
- formal research appointments;
- or guaranteed inclusion in future research or collaborative activities beyond separately established institutional processes.

C. Activity Documentation

Activity documentation shall be prepared and consolidated throughout the implementation period to support institutional learning, implementation monitoring, reporting compliance, and preservation of methodological and interdisciplinary insights generated during the program.

Activity documentation may include:

- session summaries;
- reflection outputs;
- photographs and screenshots of activities;
- presentation materials;
- conceptual workshop outputs;
- mentoring notes;
- interdisciplinary discussion summaries;
- and records of exploratory educational activities conducted during the implementation period.

The Project Team shall ensure that all documentation materials:

- remain consistent with the approved implementation framework;
- properly acknowledge participating units and approved institutional affiliations;
- and avoid misrepresentation of exploratory conceptual outputs as operational scientific products or validated systems.

Documentation of conceptual or prototype-oriented educational activities shall clearly indicate that these activities are exploratory, educational, interdisciplinary, and research-readiness-oriented.

All documentation materials shall remain subject to:

- institutional policies;
- ethical considerations;
- and applicable DOST–PCIEERD documentation and reporting requirements.

D. Financial Liquidation

Financial liquidation shall be conducted in accordance with:

- applicable DOST–PCIEERD financial guidelines;
- government accounting and auditing regulations;
- institutional financial procedures;
- and the approved Line-Item Budget (LIB) associated with the Visiting Expert Program implementation.

The Project Team and implementing institution shall ensure the proper recording, consolidation, and submission of:

- financial statements;
- disbursement records;
- official receipts;
- supporting documentation;
- and other financial requirements necessary for liquidation and audit purposes
  .

All financial transactions and expenditures shall remain limited to activities and expenses authorized under:

- the approved budget;
- the Memorandum of Agreement (MOA);
- and applicable DOST–PCIEERD policies and institutional regulations.

Any proposed modifications involving budget utilization, expenditure adjustments, or implementation-related financial concerns shall be subject to appropriate institutional and DOST–PCIEERD review and approval processes.

E. Terminal Reports

A terminal report shall be prepared at the conclusion of the implementation period to summarize the overall conduct, accomplishments, outputs, institutional learning outcomes, and implementation experiences of the Visiting Expert Program.

The terminal report may include:

- summary of completed activities;
- interdisciplinary learning outcomes;
- implementation highlights;
- synthesized methodological insights;
- participant engagement outcomes;
- documentation of capacity-building accomplishments;
- and institutional reflections related to the implementation of the program.

The terminal report shall also discuss:

- implementation challenges encountered during the program;
- operational and institutional considerations;
- and recommendations for future academic, interdisciplinary, or research-oriented initiatives related to earthquake interpretation and historical seismic analysis.

The terminal report shall not be interpreted as:

- a formal research and development completion report;
- certification of operational software systems;
- validation of analytical or prototype-oriented educational outputs;
- or endorsement of predictive seismic technologies beyond the approved scope of the Visiting Expert Program.

Any future research activities, publications, interdisciplinary initiatives, software-related developments, or extended collaborations that may emerge after the implementation period shall be pursued separately, subject to appropriate institutional review, ethical processes, authorship agreements, and applicable funding mechanisms, beyond the scope of the current VEP implementation.

All deliverables and reports identified under this section shall remain subject to the implementation boundaries, scientific limitations, governance provisions, and institutional guidelines established under these Terms of Reference and applicable DOST–PCIEERD policies.

IX. ADMINISTRATIVE ARRANGEMENTS

The administrative arrangements of the Visiting Expert Program are established to support the orderly, accountable, and institutionally consistent implementation of all activities conducted under the approved framework of the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD.

All administrative processes, coordination mechanisms, logistical activities, and documentation procedures shall remain aligned with:

- the Memorandum of Agreement (MOA);
- the approved Workplan;
- the Session Design and Outline;
- the Scheduled Plan of Activities;
- applicable institutional policies;
- and relevant DOST–PCIEERD guidelines and reporting requirements.

Administrative arrangements established under this section are intended primarily to support:

- capacity-building implementation;
- interdisciplinary learning activities;
- expert engagement;
- documentation and reporting workflows;
- and institutional coordination related to the Visiting Expert Program.

All administrative activities shall remain subject to the implementation boundaries, scientific limitations, governance provisions, ethical considerations, and institutional responsibilities established under these Terms of Reference.

A. Logistics

The implementing institution and Project Team shall provide logistical support for the proper conduct of all program activities, including online and on-site sessions, mentoring activities, workshops, interdisciplinary discussions, and synthesis sessions.

Logistical support may include:

- coordination of online meeting platforms;
- venue preparation for onsite sessions;
- scheduling support;
- participant coordination;
- preparation of session materials;
- communication support;
- attendance management;
- and other operational requirements necessary for the implementation of the program.

Logistical arrangements shall be conducted in accordance with:

- approved implementation schedules;
- institutional procedures;
- budgetary limitations;
- and applicable DOST–PCIEERD policies.

All logistical activities shall remain consistent with the approved educational and capacity-building nature of the program and shall not be interpreted as supporting operational deployment, technical implementation, or formal research and development activities beyond the approved scope of the Visiting Expert Program.

B. Communication

Official communication related to the implementation of the Visiting Expert Program shall be coordinated through the Project Leader and authorized institutional representatives in accordance with institutional and DOST–PCIEERD communication procedures.

Communication activities may involve:

- coordination with the Visiting Expert;
- dissemination of schedules and implementation updates;
- participant advisories;
- documentation requests;
- institutional coordination;
- and reporting-related communication necessary for the implementation of the program.

All communication materials, announcements, reports, and public-facing statements shall:

- accurately reflect the approved objectives and scope of the program;
- maintain scientific and institutional consistency;
- avoid unsupported claims regarding operational earthquake prediction, validated systems, or formal software deployment;
- and properly acknowledge participating institutions and approved affiliations.

The use of institutional names, affiliations, and references involving DOST–PCIEERD, DOST–PHIVOLCS, Ateneo de Davao University, and participating units shall remain subject to applicable institutional communication policies and approved implementation arrangements.

No communication activity conducted under the program shall be interpreted as constituting:

- formal institutional partnership declarations beyond the approved implementation framework;
- operational endorsement of exploratory analytical outputs;
- or public validation of conceptual or prototype-oriented educational activities conducted under the program.

C. Scheduling

The implementation schedule of the Visiting Expert Program shall follow the approved Session Design and Outline, the Scheduled Plan of Activities, and the implementation timeline reflected in the Workplan and related implementation documents.

The Project Team shall coordinate the scheduling of:

- lectures;
- workshops;
- mentoring sessions;
- interdisciplinary discussions;
- synthesis activities;
- onsite sessions;
- and related implementation activities.

Minor adjustments to scheduling, sequencing, modality, or operational timing may be implemented when necessary due to:

- institutional considerations;
- operational requirements;
- logistical constraints;
- participant availability;
- or force majeure circumstances.

Any scheduling adjustments shall remain consistent with:

- the approved objectives of the Visiting Expert Program;
- the educational and capacity-building nature of the implementation;

- and applicable institutional and DOST–PCIEERD requirements.

Scheduling modifications shall not materially alter:

- the approved implementation framework;
- the methodological direction of the program;
- or the implementation boundaries established under these Terms of Reference.

D. Honoraria

Honoraria and allowable compensation related to the implementation of the Visiting Expert Program shall be governed by:

- the approved Line-Item Budget (LIB);
- the Memorandum of Agreement (MOA);
- applicable DOST–PCIEERD financial policies;
- government accounting and auditing regulations;
- and institutional financial procedures.

Honoraria may be provided to authorized personnel, resource persons, or participating individuals in accordance with approved budget allocations and applicable institutional and government regulations.

All honoraria-related transactions shall remain subject to:

- appropriate institutional processing;
- documentation requirements;
- liquidation procedures;
- and applicable DOST–PCIEERD financial guidelines.

Honoraria provided under the Visiting Expert Program shall not be interpreted as:

- research employment arrangements;
- ownership rights over conceptual outputs;
- formal research appointments;
- or automatic entitlement to future participation in subsequent initiatives beyond separately approved institutional arrangements.

E. Documentation Procedures

Documentation procedures shall be implemented to support:

- institutional accountability;
- implementation monitoring;
- interdisciplinary learning preservation;
- reporting compliance;
- and consolidation of methodological and scientific insights generated during the program.

Documentation activities may include:

- attendance recording;
- session summaries;
- reflection outputs;
- mentoring notes;
- activity reports;
- photographs and screenshots;
- workshop documentation;
- interdisciplinary analytical summaries;
- and records of exploratory educational activities conducted during the implementation period.

The Project Team shall ensure that all documentation:

- accurately reflects the approved implementation framework;
- remains scientifically and institutionally consistent;
- properly acknowledges approved affiliations and participating units;
- and avoids misrepresentation of exploratory conceptual outputs as operational scientific systems or validated technologies.

Documentation involving conceptual, analytical, or prototype-oriented educational activities shall clearly indicate that such outputs are:

- exploratory;
- educational;
- interdisciplinary;
- and research-readiness oriented in nature.

All documentation materials generated during the implementation period shall remain subject to:

- institutional documentation policies;
- ethical considerations;
- participant governance provisions;
- and applicable DOST–PCIEERD reporting and archival requirements.

The implementing institution reserves the right to consolidate, archive, or use implementation documentation for institutional learning, reporting, academic reflection, and future planning, consistent with institutional policies and the approved scope of the Visiting Expert Program.

X. EVALUATION AND MONITORING

The evaluation and monitoring framework of the Visiting Expert Program is intended to support:

- implementation accountability;
- monitoring of program progress;
- assessment of interdisciplinary learning engagement;
- documentation of capacity-building outcomes;
- and institutional compliance with the requirements of the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD.

Evaluation and monitoring activities shall remain aligned with:

- the approved objectives of the Visiting Expert Program;
- the Memorandum of Agreement (MOA);
- the approved Workplan;
- the Session Design and Outline;
- the Scheduled Plan of Activities;
- and applicable institutional and DOST–PCIEERD implementation and reporting guidelines.

The evaluation framework shall focus primarily on:

- learning progression;
- interdisciplinary participation;
- methodological understanding;
- implementation effectiveness;
- documentation quality;
- and institutional capacity-building outcomes.

Accordingly, evaluation and monitoring activities conducted under the program shall not be interpreted as:

- validation of operational software systems;
- certification of predictive seismic technologies;
- formal technical benchmarking of exploratory analytical outputs;
- or scientific validation of conceptual or prototype-oriented educational activities developed during the implementation period.

All evaluation and monitoring activities shall remain subject to:

- scientific limitations;
- participant governance provisions;
- institutional policies;
- ethical considerations;
- and the approved implementation boundaries established under these Terms of Reference.

A. Implementation Monitoring

The Project Team shall conduct regular monitoring of program implementation to ensure that activities remain consistent with:

- approved objectives;
- implementation schedules;
- reporting requirements;
- and institutional and DOST–PCIEERD policies.

Implementation monitoring may include:

- activity tracking;
- session implementation reviews;
- participant engagement monitoring;
- attendance verification;
- documentation reviews;
- and coordination meetings involving the Project Team and relevant institutional personnel.

Monitoring activities shall focus on ensuring:

- continuity of implementation;
- interdisciplinary coherence;
- alignment with educational and methodological objectives;
- and consistency with the approved scope of the Visiting Expert Program.

B. Learning and Participation Evaluation

Participant learning and engagement may be evaluated through:

- attendance and participation records;
- reflection outputs;
- conceptual exercises;
- interdisciplinary discussions;
- workshop engagement;
- analytical synthesis activities;
- and conceptual or prototype-oriented educational presentations conducted during the implementation period.

Evaluation activities shall emphasize:

- conceptual understanding;
- interdisciplinary integration;
- uncertainty-aware scientific reasoning;
- methodological awareness;

- and responsible interpretation of earthquake-related information.

The evaluation process is intended primarily for:

- educational monitoring;
- implementation assessment;
- institutional learning;
- and research-readiness development.

Participation in evaluation activities shall not automatically establish:

- authorship rights;
- intellectual ownership;
- formal research appointments;
- or technical certification of exploratory conceptual outputs developed during the program.

C. Documentation and Reporting Review

The Project Team shall periodically review implementation documentation and reporting materials to ensure:

- completeness;
- institutional consistency;
- scientific appropriateness;
- and compliance with DOST–PCIEERD and institutional requirements.

Documentation review activities may include:

- verification of attendance records;
- review of session summaries;
- evaluation of activity documentation;
- consolidation of reflection outputs;
- and monitoring of implementation-related deliverables.

All reports and documentation materials shall:

- accurately reflect the approved implementation framework;
- avoid unsupported scientific or technical claims;
- and clearly distinguish outputs from operational or validated scientific systems.

D. Feedback and Reflection Mechanisms

The program shall incorporate feedback and reflection mechanisms intended to support:

- continuous implementation improvement;
- interdisciplinary dialogue;
- participant reflection;
- and institutional learning throughout the implementation period.

Feedback activities may include:

- participant reflections;
- session evaluations;
- mentoring feedback;
- interdisciplinary discussion reviews;
- and synthesis sessions involving conceptual and methodological reflection.

Feedback gathered during the implementation period may be used for:

- improving future capacity-building initiatives;
- refining institutional implementation practices;
- strengthening interdisciplinary engagement approaches;
- and supporting future academic and research-readiness planning beyond the scope of the current Visiting Expert Program implementation.

E. Final Evaluation and Program Synthesis

A final evaluation and synthesis activity shall be conducted near the conclusion of the implementation period to assess:

- overall implementation progress;
- interdisciplinary learning outcomes;
- participant engagement;
- methodological understanding;
- and institutional accomplishments achieved during the Visiting Expert Program.

The final evaluation process may include:

- review of compiled learning portfolios;
- synthesis discussions;
- conceptual or prototype-oriented educational presentations;
- reflection outputs;
- and assessment of interdisciplinary integration throughout the program.

The final synthesis activities are intended to consolidate:

- institutional learning;
- methodological insights;
- interdisciplinary understanding;
- and research-readiness outcomes generated during the implementation period.

The final evaluation process shall not be interpreted as:

- formal scientific validation of conceptual outputs;
- technical certification of analytical systems;
- operational evaluation of hazard technologies;
- or endorsement of predictive seismic methodologies beyond the approved scope of the Visiting Expert Program.

Any future research activities, interdisciplinary initiatives, software-related projects, or extended scholarly outputs that may emerge after the implementation period shall be pursued separately, subject to appropriate institutional review, ethical processes, authorship agreements, and applicable funding mechanisms, beyond the scope of the current VEP implementation.

All evaluation and monitoring activities identified under this section shall remain subject to the implementation limitations, scientific boundaries, governance provisions, ethical considerations, and institutional guidelines established under these Terms of Reference, as well as applicable DOST–PCIEERD policies.

XI. ETHICAL AND SCIENTIFIC CONSIDERATIONS

The Visiting Expert Program shall be implemented in accordance with principles of responsible scientific engagement, institutional accountability, interdisciplinary respect, methodological integrity, and ethical academic conduct consistent with the objectives of the Human Resource Development Program (HRDP) – Visiting Expert Program (VEP) of DOST–PCIEERD.

All activities conducted under the program shall remain aligned with:

- the approved objectives of the Visiting Expert Program;
- the Memorandum of Agreement (MOA);
- the approved Session Design and Outline;
- the Scheduled Plan of Activities;
- the approved Workplan;
- and applicable institutional, ethical, scientific, and DOST–PCIEERD policies and guidelines.

The ethical and scientific considerations established under this section are intended to support:

- responsible interpretation of earthquake-related information;
- uncertainty-aware scientific reasoning;
- proper institutional and professional conduct;
- interdisciplinary collaboration;
- and the maintenance of scientific and methodological integrity throughout the implementation of the program.

A. Responsible Scientific Interpretation

Participants, faculty members, project personnel, and collaborating individuals involved in the program shall exercise responsible scientific reasoning and interpretation in all lectures, discussions, workshops, conceptual activities, documentation processes, and exploratory analytical engagements conducted under the implementation framework.

The program recognizes that earthquake interpretation, historical seismic analysis, and intensity-based reasoning inherently involve:

- uncertainty;
- incomplete observations;
- interpretative limitations;
- observational variability;
- and context-dependent scientific judgment.

Accordingly, all discussions and analytical activities conducted under the program shall emphasize:

- methodological caution;
- interdisciplinary reasoning;
- awareness of scientific limitations;
- and responsible interpretation of earthquake-related information.

Participants and project personnel shall avoid:

- unsupported predictive claims;
- deterministic interpretation of uncertain datasets;
- misrepresentation of exploratory conceptual outputs;
- and overstatement of analytical conclusions beyond the approved educational and methodological scope of the program.

B. Scientific Scope and Limitations

The Visiting Expert Program is implemented primarily as a:

- capacity-building;
- knowledge-transfer;
- interdisciplinary learning;
- and research-readiness initiative.

The program is not intended to function as:

- an operational earthquake prediction initiative;
- a validated forecasting system;
- a deployable hazard technology project;
- or a formal research and development implementation under the Visiting Expert Program mechanism.

While the implementation framework may include conceptual discussions involving:

- computational workflows;
- geospatial systems;
- data-driven interpretation;
- visualization environments;
- exploratory analytical methods;
- artificial intelligence;
- machine learning concepts;
- and conceptual or prototype-oriented educational outputs,

Such activities shall remain:

- exploratory;
- educational;
- interdisciplinary;
- and research-readiness oriented in nature.

Any conceptual or prototype-oriented outputs generated during the implementation period shall not be interpreted as:

- operational scientific systems;
- validated analytical technologies;
- deployable software platforms;
- institutional hazard tools;
- or formally endorsed predictive methodologies.

C. Interdisciplinary and Professional Conduct

All participants, faculty members, trainees, project personnel, and collaborating individuals are expected to maintain professional and respectful conduct throughout the program.
Interdisciplinary engagement under the program shall promote:

- collaborative learning;
- respectful scientific dialogue;
- openness to differing disciplinary perspectives;
- and responsible academic interaction among participants from different fields and technical backgrounds.

Participants shall:

- respect institutional policies and implementation guidelines;
- observe appropriate professional behavior during all activities;
- contribute responsibly to discussions and collaborative exercises;
- and avoid misrepresentation of affiliations, expertise, institutional authority, or scientific interpretation.

The program recognizes that interdisciplinary activities may involve participants with varying technical backgrounds and levels of expertise. Accordingly, activities shall prioritize:

- conceptual clarity;
- constructive engagement;
- and responsible methodological discussion over technical exclusivity or disciplinary hierarchy.

D. Documentation, Attribution, and Representation

All documentation, reports, presentations, conceptual outputs, and communication materials generated during the implementation period shall:

- accurately reflect the approved scope and objectives of the program;
- maintain institutional and scientific consistency;
- and avoid unsupported technical or predictive claims.

The use of institutional names, affiliations, references, and acknowledgments involving:

- DOST–PCIEERD;
- DOST–PHIVOLCS;
- Ateneo de Davao University;
- participating academic units;
- and collaborating personnel

shall remain subject to:

- approved implementation arrangements;
- institutional communication policies;
- and applicable DOST–PCIEERD guidelines.

Participation in the program shall not automatically establish:

- research authorship;
- intellectual ownership;
- formal research appointments;
- institutional representation;
- or rights to future publication or software-related activities beyond separately established agreements and institutional processes.

Any future scholarly outputs, publications, interdisciplinary projects, software initiatives, or research activities arising after the implementation period shall be pursued separately under appropriate:

- institutional review;
- ethical processes;
- authorship agreements;
- and applicable funding or implementation mechanisms.

E. Ethical Use of Exploratory Outputs

Any conceptual, analytical, or prototype-oriented outputs developed during the implementation period shall be treated solely as outputs intended for:

- interdisciplinary learning;
- conceptual synthesis;
- methodological understanding;
- and research-readiness development.

Such outputs shall not be:

- publicly represented as operational technologies;
- used as official hazard assessment systems;
- interpreted as validated predictive tools;
- or deployed for decision-making purposes beyond the approved educational scope of the program.

The Project Team, participants, and collaborating personnel shall exercise caution in:

- discussing exploratory analytical outputs publicly;
- presenting conceptual systems outside approved contexts;
- and communicating uncertainty-related findings or interpretations derived from educational activities conducted under the program.

F. Institutional and Policy Compliance

All activities conducted under the Visiting Expert Program shall comply with:

- institutional rules and regulations;
- applicable DOST–PCIEERD policies;
- government accounting and reporting requirements;
- and relevant ethical, scientific, and administrative standards governing the implementation of the program.

The implementing institution, Project Team, Visiting Expert, and participants shall ensure that all activities remain:

- scientifically responsible;
- institutionally appropriate;
- methodologically grounded;
- and consistent with the approved implementation framework and limitations established under these Terms of Reference.

Any concerns involving:

- scientific misrepresentation;
- ethical violations;
- misuse of conceptual outputs;
- inappropriate attribution;
- or activities inconsistent with the approved implementation framework

shall be subject to appropriate institutional review and applicable administrative processes.

The ethical and scientific considerations established in this section shall apply throughout the Visiting Expert Program and to all activities, outputs, documentation, and interdisciplinary engagements conducted under the approved framework.

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